

Problem-Solution Fit

Date	28 June 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	5 Marks

Problem Solution Fit Canvas

1. CUSTOMER SEGMENT(S) [CS] - Disaster Management Authorities - Meteorological Departments - Local Government Bodies - Emergency Response Teams - NGOs involved in disaster relief - Residents in flood-prone areas - Researchers and Environmental Agencies	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited technical knowledge - Lack of real-time prediction systems - Insufficient computing resources in some regions - Dependence on manual analysis - Budget constraints	5. AVAILABLE SOLUTIONS PROS & CONS [AS] <table border="1"> <thead> <tr> <th>Current Solutions</th><th>Pros</th><th>Cons</th></tr> </thead> <tbody> <tr> <td> - Manual weather monitoring - Traditional forecasting systems - Satellite observations </td><td> - Widely used - Reliable historical data - Government-supported </td><td> - Slow analysis - Limited prediction accuracy - Requires expert interpretation - Delayed alerts </td></tr> </tbody> </table>	Current Solutions	Pros	Cons	- Manual weather monitoring - Traditional forecasting systems - Satellite observations	- Widely used - Reliable historical data - Government-supported	- Slow analysis - Limited prediction accuracy - Requires expert interpretation - Delayed alerts		
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2. PROBLEMS / PAINS + ITS FREQUENCY [PR] <table border="1"> <thead> <tr> <th>Problems</th><th>Frequency</th></tr> </thead> <tbody> <tr> <td> - Difficulty in predicting floods accurately. - Delayed disaster warnings. - Loss of lives and property due to sudden floods. - Manual analysis is time-consuming. </td><td> - During every rainy/monsoon season. - High rainfall periods. - Extreme weather events. </td></tr> </tbody> </table>	Problems	Frequency	- Difficulty in predicting floods accurately. - Delayed disaster warnings. - Loss of lives and property due to sudden floods. - Manual analysis is time-consuming.	- During every rainy/monsoon season. - High rainfall periods. - Extreme weather events.	9. PROBLEM ROOT / CAUSE [RC] - Climate change causing extreme rainfall. - Inadequate early flood prediction systems. - Poor analysis of weather data. - Lack of intelligent prediction models. - Delayed communication of flood warnings.	7. BEHAVIOR + ITS INTENSITY [BE] <table border="1"> <thead> <tr> <th>Behavior</th><th>Intensity</th></tr> </thead> <tbody> <tr> <td> - Monitor weather forecasts - Follow government alerts - Search for flood updates - Prepare emergency kits - Evacuate when warnings are issued </td><td> - High during monsoon season - Moderate throughout the year </td></tr> </tbody> </table>	Behavior	Intensity	- Monitor weather forecasts - Follow government alerts - Search for flood updates - Prepare emergency kits - Evacuate when warnings are issued	- High during monsoon season - Moderate throughout the year
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3. TRIGGERS TO ACT [TR] - Heavy rainfall forecast - Rising river or reservoir water levels - Weather department alerts - Need for timely evacuation - Government disaster preparedness initiatives	10. YOUR SOLUTION [SL] Develop an AI-powered Flood Prediction System using Machine Learning (XGBoost), Python, Flask, and IBM Cloud that: - Predicts flood probability using rainfall and weather parameters. - Provides fast and accurate flood risk prediction. - Supports disaster management authorities in early decision-making. - Reduces response time during emergencies. - Improves disaster preparedness and minimizes loss of life and property.	8. CHANNELS OF BEHAVIOR [CH] <table border="1"> <thead> <tr> <th>ONLINE</th><th>OFFLINE</th></tr> </thead> <tbody> <tr> <td> - Weather websites - Government disaster portals </td><td> - Mobile applications - Radio announcements - Local administration notices - Community awareness programs </td></tr> </tbody> </table>	ONLINE	OFFLINE	- Weather websites - Government disaster portals	- Mobile applications - Radio announcements - Local administration notices - Community awareness programs				
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